Status	Finished
Started	Thursday, 22 October 2026, 10:01 AM
Completed	Thursday, 22 October 2026, 10:05 AM
Duration	4 mins 17 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. arrange()
b. sorted()
c. sort()
d. order()

The correct answer is: sort()

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Bubble Sort
b. Merge Sort
c. Heap Sort
d. Quick Sort

The correct answer is: Bubble Sort

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Merge Sort
b. Bubble Sort
c. Insertion Sort
d. Quick Sort

The correct answer is: Merge Sort

Question **4**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By comparing elements sequentially
b. By sorting data in a single pass
c. By breaking the input into smaller parts and merging them
d. By using the bubble-up method

The correct answer is: By breaking the input into smaller parts and merging them

Question **5**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort is difficult to implement
b. Bubble Sort is less efficient for large lists
c. Bubble Sort cannot handle duplicate elements
d. Bubble Sort does not guarantee sorted order

The correct answer is: Bubble Sort is less efficient for large lists

Question **6**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It compares adjacent elements and swaps them if necessary
b. It repeatedly finds the minimum element and moves it to the sorted part of the list
c. It divides the list into two halves, sorts each half, and then merges them
d. It builds a sorted array one element at a time

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **7**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Quick sort

b. Heap sort

c. Bubble sort

d. Insertion sort

The correct answer is:
Insertion sort

Question **8**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Sorting

b. Complexity

c. Searching

d. Merging

The correct answer is:
Complexity

Question **9**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it preserves the order of all keys

b. A sorting algorithm is stable if it preserves the order of duplicate keys

c. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

d. A sorting algorithm is stable if it preserves the order of non-duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **10**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Dividing the input into parts, solving each part, and combining the solutions
- b. Sorting data sequentially
- c. Rearranging data without sorting
- d. Sorting data in a single pass

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **11**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It allows for quicker identification of duplicates
- b. It complicates the analysis process
- c. It has no effect on the analysis process
- d. It makes the analysis slower

The correct answer is: It allows for quicker identification of duplicates

Question **12**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Comparing each element with the others
- b. Dividing the input into smaller subproblems
- c. Combining sorted sublists
- d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems

Question **13**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Searching

b. Sorting

c. Rearranging

d. Merging

The correct answer is: Sorting

Question **14**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It divides the list into sublists

b. It uses the heap data structure

c. It recursively sorts subproblems

d. It bubbles up the largest element in each pass

The correct answer is: It bubbles up the largest element in each pass

Question **15**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. All of the above

b. Searching

c. Duplicates

d. Selection

The correct answer is: All of the above